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Isaac Ramírez Rojas

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Education

Instituto Técnologico San Carlos, Costa Rica

Bachelor of Science (B.S.) in Computer Engineer, GPA: 3.88/4.0 *Feb 2023 – Present*

- **Concentrations**: Artificial Intelligence, Machine Learning, Data Engineering, Database Management (SQL & NoSQL), Cloud Computing (AWS, Azure, GCP), Cybersecurity Awareness, APIs & Systems Integration, **Soft Skills (Communication, Teamwork, Leadership)**
- **Coursework**: Data Structures & Algorithms, Object-Oriented Programming, Objects & Software Design, Computer Organization, Algorithm Design, Database Systems, Statistics, Programming Languages, Probability, Project Management, Compilers & Interpreters, Software Quality Assurance, Cybersecurity, Data Analysis, Operating Systems
- **Languages**: Spanish (Native), English (Upper-Intermediate / CEFR B2 – self-assessed)
- **Honors & Certifications**: Maintained an **Honors Scholarship** throughout the entire degree by achieving a semester GPA consistently above 90/100. Completed multiple certifications in **Snowflake, Databricks, Oracle, and AWS**, demonstrating commitment to continuous learning and industry-relevant expertise.

Work Experience

Vadi Systems San Carlos, CR

AI/Software Engineering Intern (Part-Time) *Feb 2025 – Present*

- Contributed to the development of **Vadi App**, a hybrid mobile application (Angular, Capacitor, Formly) with offline-first architecture for automated property appraisals. Integrated voice-to-text, image metadata extraction, and manual inputs to streamline form completion.
- Developed **Vadi AI**, an appraisal analysis platform leveraging multiple AI models to compare valuation outcomes and generate automated recommendations on property reliability for banking and financial decision-making.

AI Research Project – Industrial Process Automation San Carlos, CR

Computer Vision Developer *August 2025 - Present*

- Developed an AI-powered system to automate time measurement in object transportation machinery, replacing manual stopwatch-based monitoring.
- Implemented computer vision models to detect operational states of objects (e.g., empty, loaded), enabling precise performance tracking and process optimization.

COMPDES 2025 – Centro Universitario de Occidente (CUNOC) Guatemala City, GT

Project Management Instructor – Oracle Primavera *July 2025*

- Delivered a professional training course on **Oracle Primavera** at the COMPDES 2025 conference, focused on project management fundamentals and applied workflows.
- Covered core topics including risk management, baseline creation, Gantt chart development, role and resource allocation, and activity scheduling for engineering and IT students at CUNOC.

Research Assistant in Artificial Intelligence San Carlos, CR

Student Software Engineer *June 2023 – Present*

- Designed and implemented advanced face anti-spoofing solutions using computer vision and deep learning, achieving reliable detection of AI-generated images, OLED screen replays, and other spoofing scenarios that bypass conventional models.
- Developed an AI-powered employee recognition system leveraging embeddings and blink-detection techniques, reducing false positives by 35% and eliminating manual clock-ins to improve organizational efficiency.

Statistical Book Created with R San Carlos, CR

Mathematics Instructor's Personal Project *August 2023 – January 2024*

- Developed proficiency in diverse ggplot2 visualization techniques in R, ensuring deeper understanding and more insightful data analysis.
- Authored and compiled a comprehensive statistical book utilizing R for advanced data analysis and visualization.
- Employed LaTeX for precise mathematical typesetting, ensuring clear and accurate presentation of formulas and equations.
- Utilized Markdown to structure and format textual content and sections, enhancing readability and organization.
- Incorporated various libraries to generate informative graphs and charts, effectively illustrating diverse data insights.

Projects

- **PuraCompra: E-commerce Intermediary Platform**: Developed an intermediary platform to streamline online purchases across Amazon, Best Buy, Micro Center, Walmart, and Gymshark; built a Node.js and Express backend with MySQL for robust database operations; created a React.js and Bootstrap frontend featuring shopping cart management, order status tracking, and user role differentiation; implemented promotion, review, and rating functionalities to enhance the shopping experience; and delivered a clean, minimalistic, accessible, and responsive design that overcomes cross-border import hurdles while providing intuitive package management and tracking.Available at: GitHub, Static website deployment for project showcase
- **Ticket Booking Server in Rust**: Developed a high-performance ticket booking server using Rust, leveraging its efficiency and stability. Designed a socket-based communication system to handle real-time ticket transactions and seat availability updates. Implemented an algorithm to determine optimal seating based on visibility percentage and proximity to desired locations. Ensured robust error handling and concurrency management to maintain server reliability under high load. Available at: GitHub
- **iFixIt**: Developed iFixIt, a web and mobile platform for automotive workshop management enabling administrators to register vehicles by license plate, log repair details (photos, audio notes, and descriptions) organized by date, and set service pricing; and empowering customers via the mobile app to track order progress, view performed work, locate nearby workshops with Google Maps integration, and schedule vehicle pickup. Available at: GitHub **Live Demo:** Static website deployment for project showcase
- **Web Scraping App**: Tackled the tedious process of manually hunting for game promotions by creating an automated Scrapy crawler to extract pricing data from G2A, Instant Gaming, and PS4 Store; stored results in a JSON-based mini-database for instant lookups; built a Flask backend to serve up-to-date pricing APIs; and designed a Vue.js frontend for a responsive, user-friendly interface with automatic updates. Available on GitHub: Web-Scraping-Game-Crawler.
- **Statistical Book Created with R**: Authored a comprehensive statistical book that integrates advanced data analysis and visualization techniques using R. Combined LaTeX for mathematical notation, Markdown for structured content, and various R libraries to generate high-quality statistical graphs. The book serves as a resource for students and researchers, offering in-depth insights into statistical methodologies. Available at: Statistical Book.
- **Research Assistant in Artificial Intelligence**: Developed a software solution for detecting real versus fake individuals using Face Anti-Spoofing techniques. Trained YOLO models to recognize various types of spoofed images related to facial authentication and object detection. Implemented eye detection algorithms to verify authenticity by analyzing eye blinks. Additionally, conducted research on Vision Transformer architectures for enhanced image recognition capabilities. More details at: AI Research Project, AI Face Detection

Certifications

Snowflake

- Hands-On Essentials: Data Engineering Workshop – Fundamentals of data engineering on Snowflake
- Hands-On Essentials: Data Science Workshop – Applied data science workflows on Snowflake
- Hands-On Essentials: Collaboration, Marketplace & Cost Estimation Workshop – Data sharing and cost optimization
- Hands-On Essentials: Data Application Builders Workshop – Building data-driven applications
- Hands-On Essentials: Data Warehousing Workshop – Data analytics and warehousing best practices

Databricks

- Academy Accreditation – Databricks Fundamentals (valid through Aug 2026): core skills in data processing, Delta Lake, and collaborative analytics

Oracle

- Primavera Cloud Schedule Management – Project planning, risk management, scheduling, and resource allocation

Amazon Web Services (AWS)

- AWS Cloud Practitioner Essentials – Core AWS services, architecture, and cloud security
- AWS Educate: Introduction to Generative AI – Foundations of generative AI applications
- AWS Educate: Job Roles in the Cloud – Cloud career paths and practical skills
- AWS Educate: Machine Learning Foundations – Introductory ML principles and AWS implementation

Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C++, C, Rust, R, Go, LaTeX
Version Control & Collaboration: Git, GitHub, GitLab
Web Development: Node.js, React, Vite, Angular, HTML/CSS, TailwindCSS, Material Tailwind, Formly
Mobile Development: Capacitor (Hybrid Apps), Firebase
Databases: PostgreSQL, MySQL, MongoDB, Supabase
Backend Frameworks: Express.js, Flask, .NET (basic experience)
Cloud Platforms: AWS, Azure, Google Cloud, Render, Heroku, Vercel
Cloud Services: AWS Cognito, IAM, CloudWatch
Containerization & Virtualization: Docker
Operating Systems: Linux (Fedora, Debian-based, Arch-based), macOS, Windows
Data Science & Machine Learning: PyTorch, TensorFlow, OpenCV, scikit-learn
Data Engineering & Big Data: Snowflake, Databricks, SQL (joins, triggers, views, functions), ETL Pipelines
Project Management: Oracle Primavera, Agile Methodologies (Scrum, Extreme Programming, Kanban, Waterfall)
Documentation: Markdown, LaTeX, Typst, Pandoc
Productivity Tools: Notion, Jira, Trello

Extracurricular Activities

Hardware Enthusiast: Passionate about building and customizing technology as a hobby. Specialized in custom keyboard assembly and custom PC builds, including advanced liquid cooling systems. Experienced in modifications, maintenance, and upgrades across various devices, with a broad knowledge of emerging technologies and future-oriented vision.
Data Analysis Enthusiast: Recently discovered a passion for data analysis through SQL. Actively learning and practicing complex queries, joins, functions, triggers, and views to gain deeper insights from data and strengthen database management skills.
Self-Learner & Explorer: Highly curious and self-driven, with a strong passion for exploring unfamiliar topics to deeply understand their purpose, real-world applications, and implementation. Continuously questions existing approaches, fostering critical thinking and innovative problem-solving.
Linux Enthusiast: Curious by nature, with extensive hands-on experience testing multiple Linux distributions including Fedora, Debian-based, and Arch-based systems (Arch, Manjaro, EndeavourOS). Strong preference for Arch-based environments due to their flexibility and performance. Daily user of Neovim, enjoying customization and efficiency in development workflows.

References

Ph.D. Abel Méndez Porras: *Data Processing, Image Processing & Artificial Intelligence*
Instituto Tecnológico de Costa Rica
amendez@tec.ac.cr
Jorge Alfaro Velasco: *Professor*
Instituto Tecnológico de Costa Rica
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